Discrete Mathematics Homework #1

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Question No 17

Chapter No x Last 3 digit of my num
(1 x 017) mod 35 = 17

Question

Show that the argument with piremises "The tooth fairy is a real person" and "The tooth fairy is not a real person" and conclusion "You can find gold at the end of the rainbow" is a valid argument. Does this show that the conclusion is true?

Answer

The argument is like $p \wedge p' \equiv q$, so for q be true p and p' be true at the same time. It is impossible.

in this situation

p	p'	q
1	0	0
0	1	0

Question No 34

Chapter No x Last 3 digit of my num
(2 x 017) mod 35 = 34

Question

Show that the set of all finite subsets of the set of positive integers is a countable set.

Answer

Let X be the set of all finite subsets of the set of positive integers z^+.
Let Y_n be the set of all finite subsets of the set of positive integers z^+ for which the sum of all elements in the subsets is n.

Since n is finite, there are a finite number of ways to obtain a sum of n out of positive integers and thus Y_n is a infinite set as well.

Since Y_n infinite, we can list all sets in Y_n.

Next, we can list the sets of X as first listing all the sets of Y_1, then listing all the sets of Y_2 and so on. Since it is possible to list all the sets of A, A is countable.

The set of all finite subsets of the set of positive integers is a countable set.

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Question 16

Chapter No x Last 3 digit of my num
(3 x 017) mod 35 = 16

Question

Show that $8x^2+12x+100\log x$ is $O(x^2)$.

Answer

When $x>0$: $\log x \leq x$

let us choose $k=12$. For $x>12$, we then obtain:

$$\begin{aligned} |8x^2+12x+100\log x| &= 8x^2+12x+100\log x \\ &= 8x^2+12.1.x+10.10.\log x \\ &< 8x^2+x.x.x+x.x.\log x \\ &= 8x^2+x^2+x^2 \\ &= 10x^2 \\ &= 10|x^2| \end{aligned}$$

Definition

Big-O Notation: $f(x)$ is $O(g(x))$ if there exists constants C and k such that $|f(x)| \leq C\,|g(x)|$ whenever $x>k$.

C then needs to be chosen as at least 10, thus let us choose $C=10$.

By the definition of the Big-O notation, we then obtain that $8x^2+12x+100\log x$ is $O(x^2)$ with $k=12$ and $C=10$.

As I showed $8x^2+12x+100\log x$ is $O(x^2)$

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Question No 33

Chapter No × Last 3 digits of my num
(4 × 017) mod 35 = 33

Question

Determine whether the integers in each of these sets are mutually relatively prime.

a) 8, 10, 12
b) 12, 15, 25
c) 15, 21, 28
d) 21, 24, 28, 32

Answer

a) Since 2 is a factor of all three of these integers, this set is not mutually relatively prime.

b) Since 12 and 25 share no common factors, this set has greatest common divisor 1, so it is mutually relatively prime. Because if two of the integers in a set are relatively prime, then the set is automatically mutually relatively prime.

c) Since 15 and 28 share no common factors, this set has greatest common divisor 1, so it is mutually relatively prime.

d) Since 21 and 32 share no common factors, this set has greatest common divisor 1, so it is mutually relatively prime.

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Question 32

Chapter No x Last 3 digits of my num

$(6 \times 017) \bmod 35 = 32$

Question

We will count the number of triples (i,j,k) where i, j, and k are integers such that $0 \leq i < k$, $0 \leq j < k$, and $1 \leq k \leq n$ in two ways.

a) Show that there are k^2 such triples with a fixed k. Deduce that there are $\sum_{k=1}^{n} k^2$ such triples.

Answer for a

To count the no. of triples (i,j,k) such that $0 \leq i < k$, $0 \leq j < k$ and $1 \leq k \leq n$, one must observe that if we fix the integer k; then the nos of options for both i and j are k, as the options are one of the elements of the set $\{0,1,\ldots,k-1\}$, thus making the no. of different combinations of the pair (i,j) equal to k^2. As k can be anything between 1 to n, the total no. of such triples is

$$\sum_{k=1}^{n} k^2.$$

b) Show that the no. of such triples with $0 \leq i < j < k$ and the number of such triples with $0 \leq j < i < k$ both equal $C(n+1,3)$.

Answer for b

$0 \leq i < j < k$ must be same as the no. of three-element subsets of an $(n+1)$-set which is $C(n+1,3)$. Cause the range of the numbers now starts from 0 (instead of 1) upto n.

The case when $0 \leq j < i < k$ is symmetric and thus, has the same no. of options for such triples.

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Question 32

Chapter NO × Last 3 digits of my num

$(6 \times 017) \bmod 35 = 32$

Question

We will count the number of triples (i,j,k) where i, j and k are integers such that $0 \le i < k$, $0 \le j < k$ and $1 \le k \le n$ in two ways.

c) Show that the number of such triples with $0 \le i = j < k$ equals $C(n+1,2)$.

Answer for c

The only remaining sub-case is when $0 \le i = j < n$. Observe that we have to choose only two integers out of the possible $n+1$ options because whenever i is chosen, j is chosen simultaneously. Thus the total number of ways in which such triples can be chosen is $C(n+1,2)$

d) Combining part (a) with part (b) and part (c), conclude that $\sum_{k=1}^{n} k^2 = 2C(n+1,3) + C(n+1,2)$

$= n(n+1)(2n+1)/6$

Answer for d

We can equate the no. of ways in which a triple can be chosen from the specified range $\{0, 1, \dots n\}$ obtained using two different approaches.

$$\sum_{k=1}^{n} k^2 = 2C(n+1,3) + C(n+1,2) = 2\frac{n(n+1)(n-1)}{3!} + \frac{n(n+1)}{2!} = \frac{n(n+1)(2n+1)}{6}$$